```
****************************************************
*                                                  *
*           STAAD.Pro                              *
*           Version  2007    Build 04              *
*           Proprietary Program of                 *
*           Research Engineers,  Intl.             *
*           Date=    OCT 19, 2012                  *
*           Time=    10:53:22                      *
*                                                  *
*      USER ID: Parsons Binckerhoff                *
****************************************************
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```
     1. STAAD SPACE
INPUT FILE: 121009-Canopy_1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 05-MAY-10
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 7.044 0 0; 4 4.696 0 0; 5 0 1.4 0; 6 0 0 6; 7 4.696 0 6
     9. 8 7.044 0 6; 9 0 1.4 6; 10 5.871 0 6; 11 5.871 0 0; 12 3.523 0 0; 13 3.523 0 6
    10. 14 2.35 0 0; 15 2.35 0 6; 16 1.177 0 0; 17 1.177 0 6; 18 7.044 0 4.8
    11. 19 7.044 0 3.6; 20 7.044 0 2.4; 21 7.044 0 1.2; 22 5.871 0 4.8; 23 5.871 0 3.6
    12. 24 5.871 0 2.4; 25 5.871 0 1.2; 26 4.696 0 1.2; 27 4.696 0 2.4; 28 4.696 0 3.6
    13. 29 4.696 0 4.8; 30 3.523 0 1.2; 31 3.523 0 2.4; 32 3.523 0 3.6; 33 3.523 0 4.8
    14. 34 2.35 0 1.2; 35 2.35 0 2.4; 36 2.35 0 3.6; 37 2.35 0 4.8; 38 1.177 0 1.2
    15. 39 1.177 0 2.4; 40 1.177 0 3.6; 41 1.177 0 4.8; 42 0.0770004 0 0
    16. 43 0.0770004 0 1.2; 44 0.0770004 0 2.4; 45 0.0770004 0 3.6; 46 0.0770004 0 4.8
    17. 47 0.0770004 0 6; 48 3.973 0 0; 49 3.973 0 6; 50 7.044 0 -0.8; 51 4.696 0 -0.8
    18. 52 5.871 0 -0.8; 53 3.523 0 -0.8; 54 2.35 0 -0.8; 55 1.177 0 -0.8
    19. 56 0.0770004 0 -0.8; 57 4.696 0 6.8; 58 7.044 0 6.8; 59 5.871 0 6.8
    20. 60 3.523 0 6.8; 61 2.35 0 6.8; 62 1.177 0 6.8; 63 0.0770004 0 6.8
    21. MEMBER INCIDENCES
    22. 1 1 42; 3 4 11; 5 6 47; 6 7 10; 8 8 18; 9 10 8; 10 11 2; 11 10 22; 12 4 26
    23. 13 12 48; 14 13 49; 15 12 30; 16 14 12; 17 15 13; 18 14 34; 19 16 14; 20 17 15
    24. 21 16 38; 22 18 19; 23 19 20; 24 20 21; 25 21 2; 26 22 23; 27 23 24; 28 24 25
    25. 29 25 11; 30 26 27; 31 27 28; 32 28 29; 33 29 7; 34 30 31; 35 31 32; 36 32 33
    26. 37 33 13; 38 34 35; 39 35 36; 40 36 37; 41 37 15; 42 38 39; 43 39 40; 44 40 41
    27. 45 41 17; 46 38 34; 47 39 35; 48 40 36; 49 41 37; 50 42 16; 51 42 43; 52 43 44
    28. 53 44 45; 54 45 46; 55 47 17; 56 46 47; 57 43 38; 58 44 39; 59 45 40; 60 46 41
    29. 61 25 21; 62 26 25; 63 30 26; 64 34 30; 65 24 20; 66 27 24; 67 31 27; 68 35 31
    30. 69 23 19; 70 28 23; 71 32 28; 72 36 32; 73 22 18; 74 29 22; 75 33 29; 76 37 33
    31. 77 48 4; 78 49 7; 79 5 48; 80 9 49; 81 56 42; 82 55 16; 83 54 14; 84 53 12
    32. 85 51 4; 86 52 11; 87 50 2; 88 47 63; 89 17 62; 90 15 61; 91 13 60; 92 7 57
    33. 93 10 59; 94 8 58; 95 63 62; 96 56 55; 97 59 58; 98 57 59; 99 60 57; 100 61 60
    34. 101 62 61; 102 52 50; 103 51 52; 104 53 51; 105 54 53; 106 55 54
    35. DEFINE MATERIAL START
    36. ISOTROPIC STEEL
    37. E 2.05E+008
    38. POISSON 0.3
    39. DENSITY 25.12
    40. ALPHA 1.2E-005
    41. DAMP 0.03
    42. END DEFINE MATERIAL
    43. CONSTANTS
    44. MATERIAL STEEL ALL
    45. MEMBER PROPERTY BRITISH
    46. 79 80 TABLE ST PIP1145.0
    47. 8 11 12 15 18 21 TO 49 51 TO 54 56 TO 76 81 TO 106 TABLE ST TUB1501505.0
    48. MEMBER PROPERTY JAPANESE
    49. 1 3 5 6 9 10 13 14 16 17 19 20 50 55 77 78 TABLE ST TUBE TH 0.006 WT 0.2 DT 0.4
    50. SUPPORTS
    51. 5 9 PINNED
    52. 1 6 FIXED BUT KMX 0.001 KMY 0.001 KMZ 0.001
    53. MEMBER RELEASE
    54. 79 80 95 TO 106 END MY MZ
    55. 46 TO 49 57 TO 76 95 TO 106 START MY MZ
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46 TO 49 57 TO 76 END MY MZ
LOAD 1 LOADTYPE NONE  TITLE DL
MEMBER LOAD
11 12 15 18 21 26 TO 45 82 TO 86 89 TO 93 UNI GY -0.59
8 22 TO 25 51 TO 54 56 81 87 88 94 UNI GY -0.3
SELFWEIGHT Y -1 LIST 1 3 5 6 8 TO 106
LOAD 2 LOADTYPE NONE  TITLE LL
MEMBER LOAD
11 12 15 18 21 26 TO 45 82 TO 86 89 TO 93 UNI GY -0.59
8 22 TO 25 51 TO 54 56 81 87 88 94 UNI GY -0.3
LOAD 3 LOADTYPE NONE  TITLE DL NOTIONAL Z-DIRECTION
NOTIONAL LOAD
1 Z 0.005
LOAD 4 LOADTYPE NONE  TITLE LL NOTIONAL Z-DIRECTION
NOTIONAL LOAD
2 Z 0.005
LOAD 5 LOADTYPE WIND  TITLE WIND Y-DIRECTION LEEWARD
MEMBER LOAD
11 12 15 18 21 26 TO 45 82 TO 86 89 TO 93 UNI GY -0.482
8 22 TO 25 51 TO 54 56 81 87 88 94 UNI GY -0.241
LOAD 6 LOADTYPE WIND  TITLE DL NOTIONAL X-DIRECTION
NOTIONAL LOAD
1 X 0.005
LOAD 7 LOADTYPE WIND  TITLE WIND Y-DIRECTION SUCTION
MEMBER LOAD
11 12 15 18 21 26 TO 45 82 TO 86 89 TO 93 UNI GY 1.2
8 22 TO 25 51 TO 54 56 81 87 88 94 UNI GY 0.6
LOAD 8 LOADTYPE WIND  TITLE LL NOTIONAL X-DIRECTION
NOTIONAL LOAD
2 X 0.005
LOAD COMB 9 1.0DL
1 1.0
LOAD COMB 10 1.0DL+1.0LL
1 1.0 2 1.0
LOAD COMB 11 1.0 ( DL+LL ) + Z
1 1.0 2 1.0 3 1.0 4 1.0
LOAD COMB 12 1.0 ( DL+LL ) - Z
1 1.0 2 1.0 3 -1.0 4 -1.0
LOAD COMB 13 1.0DL+1.0WL3
1 1.0 5 1.0
LOAD COMB 14 1.0 ( DL+LL ) + X
1 1.0 2 1.0 6 1.0 8 1.0
LOAD COMB 15 1.0DL+1.0WL5
1 1.0 7 1.0
LOAD COMB 16 1.0 ( DL+LL ) - X
1 1.0 2 1.0 6 -1.0 8 -1.0
LOAD COMB 17 1.4 DL+ 1.6 LL + Z(FACTORED)
1 1.4 2 1.6 3 1.4 4 1.6
LOAD COMB 18 1.4 DL+ 1.6 LL - Z(FACTORED)
1 1.4 2 1.6 3 -1.4 4 -1.6
LOAD COMB 19 1.0DL+1.0LL+1.0WL3
1 1.0 2 1.0 5 1.0
LOAD COMB 20 1.4 DL+ 1.6 LL + X(FACTORED)
1 1.4 2 1.6 6 1.4 8 1.6
LOAD COMB 21 1.0DL+1.0LL+1.0WL5
1 1.0 2 1.0 7 1.0
LOAD COMB 22 1.4 DL+ 1.6 LL - X(FACTORED)
1 1.4 2 1.6 6 -1.4 8 -1.6
LOAD COMB 23 1.4DL
1 1.4
LOAD COMB 24 1.4DL+1.6LL
1 1.4 2 1.6
LOAD COMB 27 1.2DL+1.2LL+1.2WL3
1 1.2 2 1.2 5 1.2
LOAD COMB 29 1.2DL+1.2LL+1.2WL5
1 1.2 2 1.2 7 1.2
LOAD COMB 33 1.4DL+1.4WL3
1 1.4 5 1.4
LOAD COMB 35 1.4DL+1.4WL5
1 1.4 7 1.4
PERFORM ANALYSIS
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        P R O B L E M   S T A T I S T I C S
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    NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =    62/   103/     4

          SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

    ORIGINAL/FINAL BAND-WIDTH=    50/     8/      54 DOF
    TOTAL PRIMARY LOAD CASES =    8, TOTAL DEGREES OF FREEDOM =    360
    SIZE OF STIFFNESS MATRIX =       20 DOUBLE  KILO-WORDS
    REQRD/AVAIL. DISK SPACE  =     12.4/      0.0 MB



  **** WARNING : AVAILABLE HARD DISK SPACE MAY NOT BE
ENOUGH TO COMPLETE EXECUTION. IF YOUR AVAILABLE HARD DISK
SPACE ON THE ANALYSIS DRIVE IS GREATER THAN 3GB THIS MESSAGE
MAY BE ERRONEOUS



  127. DEFINE ENVELOP
  128. 9 TO 16 19 21 ENVELOP 1 TYPE SERVICEABILITY
  129. 17 18 20 22 TO 24 27 29 33 35 ENVELOP 2 TYPE STRESS
  130. END DEFINE ENVELOP
  131. PARAMETER 1
  132. CODE BS5950
  133. UNL 6 MEMB 8 11 12 15 18 21 TO 45 51 TO 54 56
  134. UNL 7.4 MEMB 1 3 5 6 9 10 13 14 16 17 19 20 50 55 77 78
  135. MAIN 1 MEMB 1 3 5 6 9 10 13 14 16 17 19 20 50 55 77 78
  136. CHECK CODE ALL

                        STAAD.Pro CODE CHECKING - (BSI )
                        ***********************

 PROGRAM CODE REVISION V2.12_5950-1_2000



ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE         RESULT/   CRITICAL COND/      RATIO/     LOADING/
                           FX            MY             MZ        LOCATION
=======================================================================


      1 ST   TUB E         PASS     BS-4.8.3.2         0.091        27
                         129.36 C        -0.06         -0.44         0.08
      3 ST   TUB E         PASS     BS-4.8.2.2         0.101        27
                           3.26 T         0.15        -20.37         0.00
      5 ST   TUB E         PASS     BS-4.8.3.2         0.091        27
                         129.36 C         0.06         -0.44         0.08
      6 ST   TUB E         PASS     BS-4.8.2.2         0.101        27
                           3.26 T        -0.15        -20.37         0.00
      8 ST TUB1501505.0    PASS     ANNEX I.1          0.113        27
                           0.28 C         0.53          4.31          -
      9 ST   TUB E         PASS     BS-4.8.2.2         0.027        27
                           1.59 T        -0.13         -5.21         0.00
     10 ST   TUB E         PASS     BS-4.8.2.2         0.027        27
                           1.59 T         0.13         -5.21         0.00
     11 ST TUB1501505.0    PASS     BS-4.9             0.129        27
                           0.00 T         0.53          4.99         0.00
     12 ST TUB1501505.0    PASS     BS-4.9             0.127        27
                           0.00 T         0.57          4.87         0.00
     13 ST   TUB E         PASS     BS-4.8.3.3.1       0.257        27
                         124.29 C         0.11         35.78          -
     14 ST   TUB E         PASS     BS-4.8.3.3.1       0.257        27
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